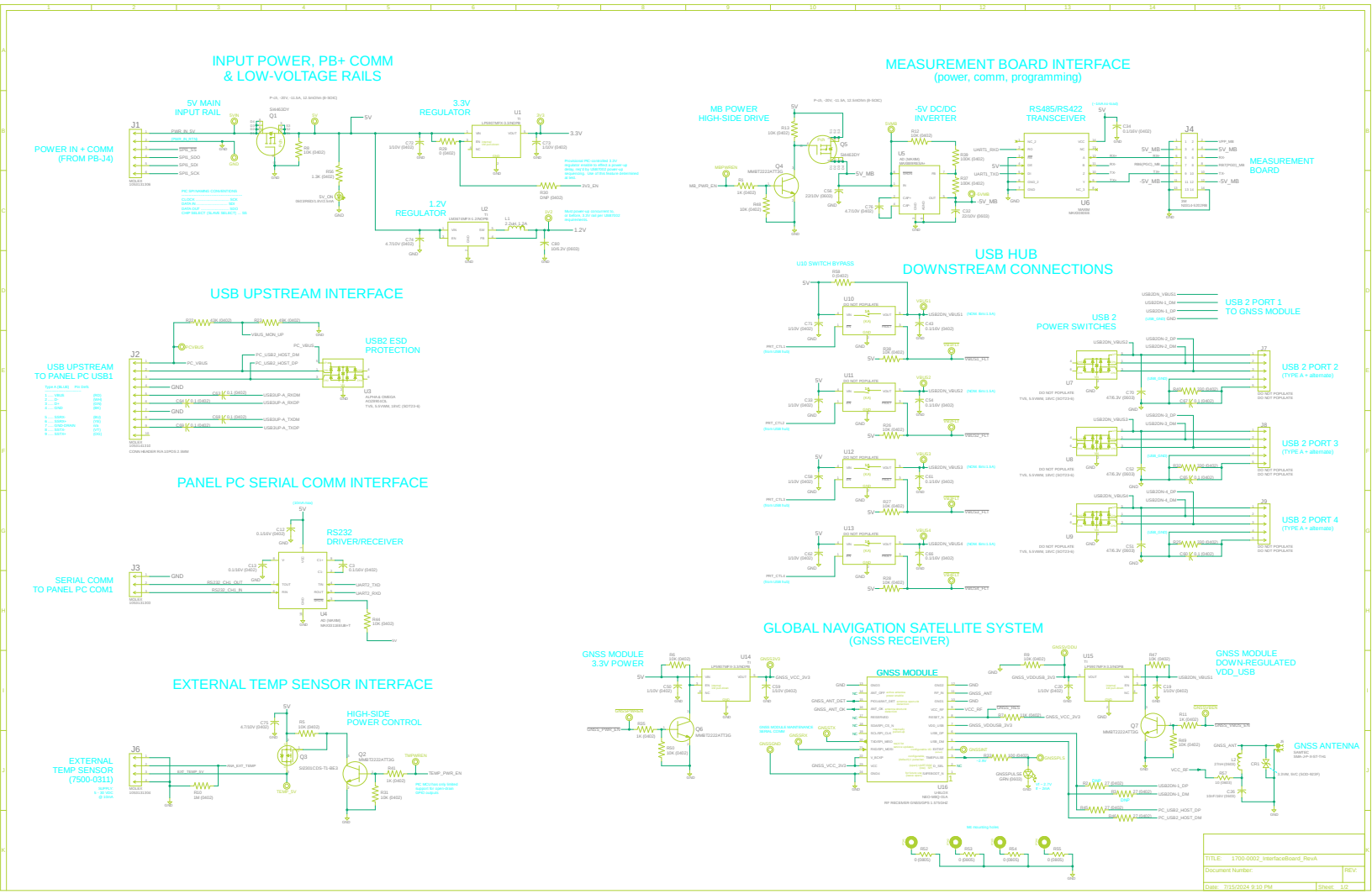
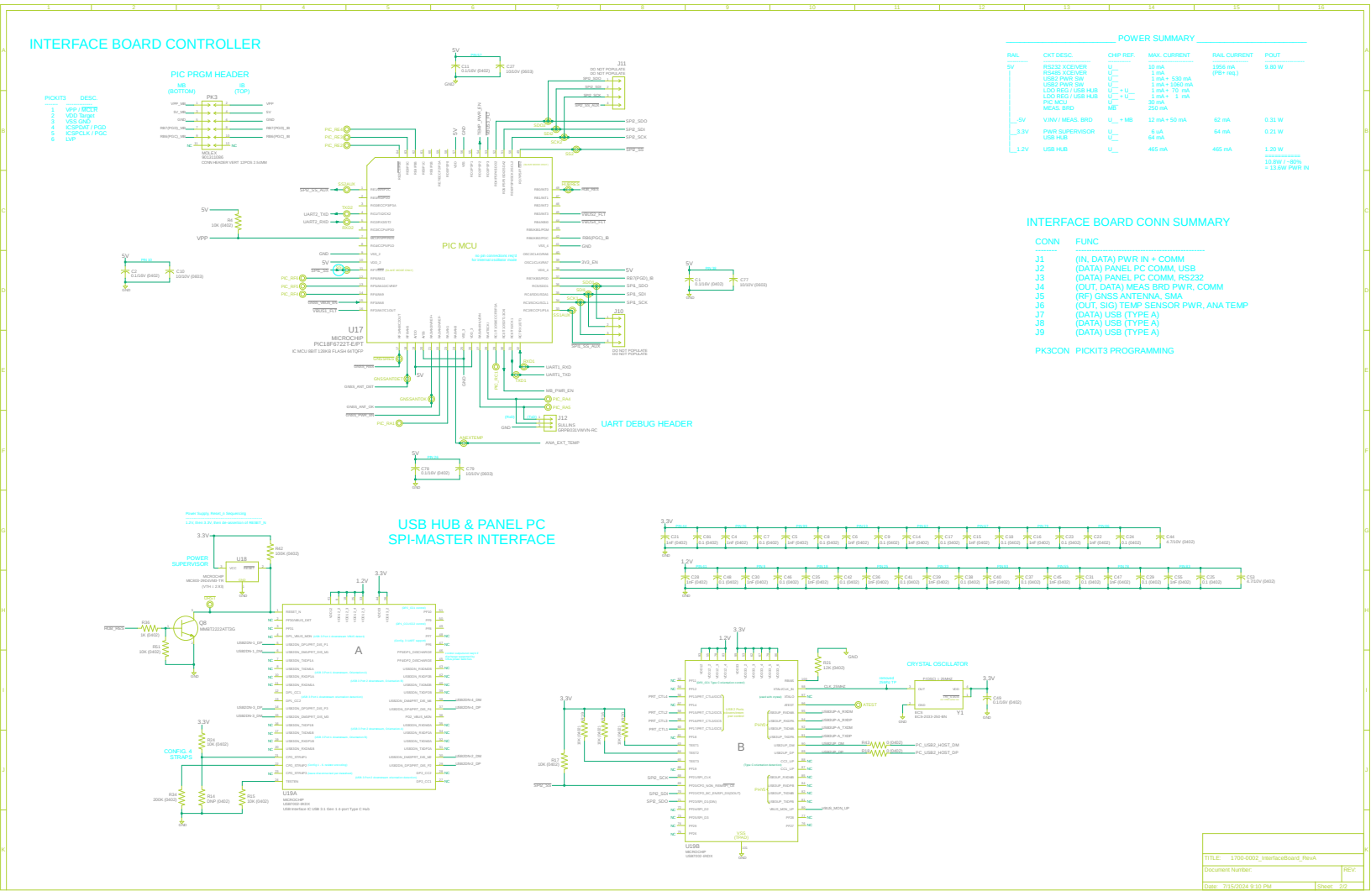

INPUT POWER, PB+ COMM
& LOW-VOLTAGE RAILS
POWER IN + COMM
(FROM PB-J4)
5V MAIN
INPUT RAIL
3.3V
REGULATOR
1.2V
REGULATOR
MEASUREMENT BOARD INTERFACE
(power, comm, programming)
MB POWER
HIGH-SIDE DRIVE
-5V DC/DC
INVERTER
RS485/RS422
TRANSCEIVER
MEASUREMENT
BOARD
USB HUB
DOWNSTREAM CONNECTIONS
USB 2
POWER SWITCHES
USB 2 PORT 1
TO GNSS MODULE
USB 2 PORT 2
(TYPE A + alternate)
USB 2 PORT 3
(TYPE A + alternate)
USB 2 PORT 4
(TYPE A + alternate)
USB UPSTREAM INTERFACE
USB UPSTREAM
TO PANEL PC USB1
USB2 ESD
PROTECTION
PANEL PC SERIAL COMM INTERFACE
SERIAL COMM
TO PANEL PC COM1
RS232
DRIVER/RECEIVER
EXTERNAL TEMP SENSOR INTERFACE
EXTERNAL
TEMP SENSOR
(7500-0311)
HIGH-SIDE
POWER CONTROL
GLOBAL NAVIGATION SATELLITE SYSTEM
(GNSS RECEIVER)
GNSS MODULE
3.3V POWER
GNSS MODULE
DOWN-REGULATED
VDD_USB
GNSS ANTENNA
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INTERFACE BOARD CONTROLLER
PIC PRGM HEADER
MB (BOTTOM)
IB (TOP)
PICKIT3 DESC.
VPP / MCLR
VDD Target
VSS GND
ICSPDAT / PGD
ICSPCLK / PGC
LVP
PIC MCU
U17
MICROCHIP
PIC18F6722T-E/PT
UART DEBUG HEADER
POWER SUMMARY
RAIL
CKT DESC.
CHIP REF.
MAX. CURRENT
RAIL CURRENT
POUT
5V
RS232 XCEIVER
RS485 XCEIVER
USB2 PWR SW
USB2 PWR SW
LDO REG / USB HUB
LDO REG / USB HUB
PIC MCU
MEAS. BRD
10 mA
1 mA
1 mA + 530 mA
1 mA + 1060 mA
1 mA + 70 mA
1 mA + 1 mA
30 mA
250 mA
1956 mA
(PB+ req.)
9.80 W
3.3V
PWR SUPERVISOR
USB HUB
6 uA
64 mA
64 mA
0.21 W
1.2V
USB HUB
465 mA
465 mA
1.20 W
10.8W / ~80%
= 13.8W PWR IN
INTERFACE BOARD CONN SUMMARY
CONN
FUNC
J1
(IN, DATA) PWR IN + COMM
J2
(DATA) PANEL PC COMM, USB
J3
(DATA) PANEL PC COMM, RS232
J4
(OUT, DATA) MEAS BRD PWR, COMM
J5
(RF) GNSS ANTENNA, SMA
J6
(OUT, SIG) TEMP SENSOR PWR, ANA TEMP
J7
(DATA) USB (TYPE A)
J8
(DATA) USB (TYPE A)
J9
(DATA) USB (TYPE A)
PK3CON PICKIT3 PROGRAMMING
USB HUB & PANEL PC
SPI-MASTER INTERFACE
POWER SUPERVISOR
CONFIG_4 STRAPS
CRYSTAL OSCILLATOR
A
B
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